Building quantum codes with transversal CCZ

from projective geometry

Bohan Lu, Duke University

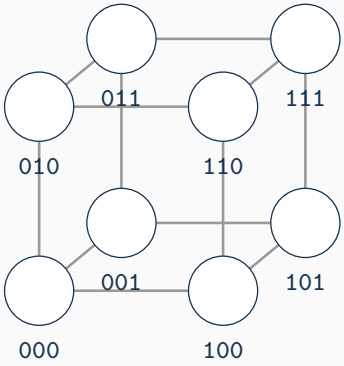
Joint work with Ken Brown

Virginia Tech Applied Algebra Seminar

October 22, 2026

arXiv:2610.09341

The cube code $[\![8, 3, 2]\!]$

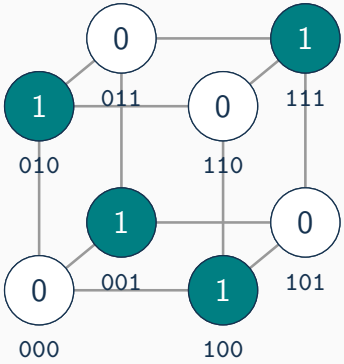


Qubits are the points $u \in \mathbb{F}_2^3$. A binary word is a function $c\colon \mathbb{F}_2^3 \to \mathbb{F}_2$, with $c \wedge c' = cc'$ and $\mathrm{wt}(c) = |c^{-1}(1)|$. $\mathrm{RM}(r, m)$ is the evaluation code of polynomials of degree $\leq r$ on $\mathbb{F}_2^m$.

A CSS code [CS96, Ste96] is given by $\mathcal{C}_2 = \mathrm{rowspan}\, S \subset \mathcal{C}_1 = \mathrm{rowspan}\, [\begin{smallmatrix} K \\ S \end{smallmatrix}]$, with $|x\rangle_L \propto \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle$ for $x \in \mathbb{F}_2^k$, where $|c\rangle$ is the basis state labelled by c. X checks span $\mathcal{C}_2$, Z checks span $\mathcal{C}_1^\perp$, and $Z_u|c\rangle = (-1)^{c(u)}|c\rangle$.

[CS96] A. R. Calderbank and P. W. Shor, Good quantum error-correcting codes exist, Phys. Rev. A 54, 1098 (1996).
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Values of $u_1 \oplus u_2 \oplus u_3$.

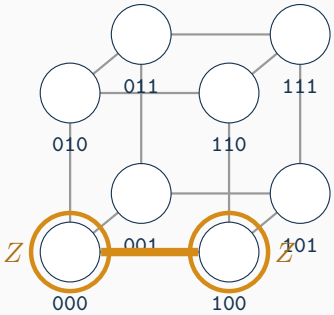
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Cube: $S = \mathbf{1}$, $K = (u_1; u_2; u_3)$, so $\mathcal{C}_1 = \mathrm{RM}(1, 3)$ and $|x\rangle_L = \big(|x \cdot u\rangle + |1 \oplus x \cdot u\rangle\big)/\sqrt{2}$.

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The error $Z_{000}Z_{100}$.

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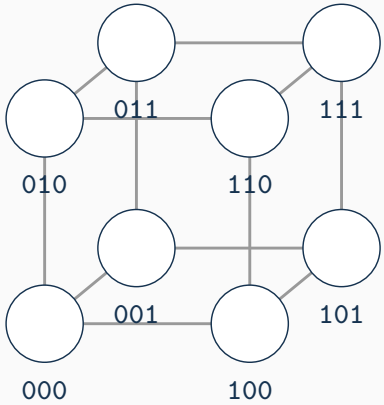
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$Z_{000}Z_{100}$ commutes with $X^{\otimes 8}$ and acts as $(-1)^{x_1}$, so $d = 2$.

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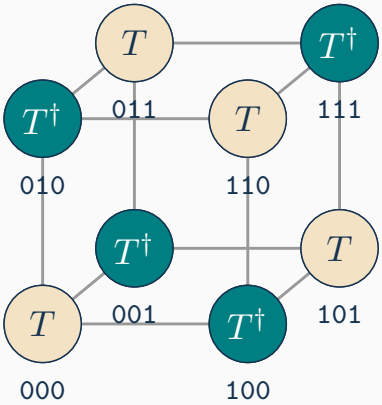
A transversal $T/T^\dagger$ layer gives logical CCZ on the cube



$T = \mathrm{diag}(1, e^{i\pi/4})$ and $\mathrm{CCZ}|x\rangle_L = (-1)^{x_1 x_2 x_3}|x\rangle_L$. Both are non-Clifford, and Clifford gates alone are not universal. A *transversal* layer is a tensor product of one-qubit gates, so one gate fault affects one qubit.

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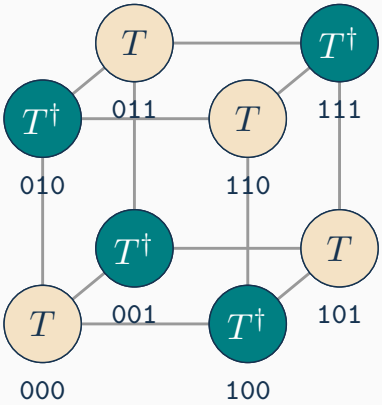
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With $\Gamma_u = (-1)^{p(u)}$, the layer multiplies $|c\rangle$ by $e^{i\pi W_\Gamma(c)/4}$, where

$$W_\Gamma(c) = \sum_u \Gamma_u \, c(u) = d_H(p, c) - 4.$$

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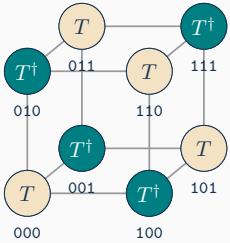
$$W_\Gamma(c) = \sum_u \Gamma_u \, c(u) = d_H(p, c) - 4.$$

For $x \neq 111$, $p \oplus c$ is nonconstant affine, so $W_\Gamma(c) = 0$.
For $x = 111$, $c \in \{p, 1 \oplus p\}$ and $W_\Gamma(c) = \mp 4$, phase -1.

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Weights of sums expand into overlaps

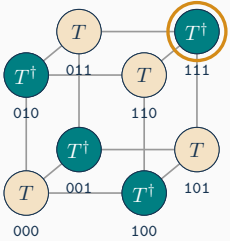
$$\mathrm{wt}(a_1 \oplus \cdots \oplus a_m) = \sum_{\emptyset \neq J \subseteq \{1,\ldots,m\}} (-2)^{|J|-1}\ \mathrm{wt}\Big(\bigwedge_{j \in J} a_j\Big).$$



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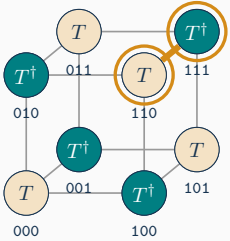


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$K_1 \wedge K_2 = u_1 u_2$ is an edge with one T and one $T^\dagger$: $W_\Gamma = 0$.

The gate condition is a congruence on generator rows

A layer $\Gamma \in \{\pm 1\}^n$ implements logical CCZ if and only if

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8 \qquad \text{for all } x \in \mathbb{F}_2^3,\ y \in \mathbb{F}_2^s.$$

The phase depends on W_Γ mod 8. Independence of y gives every word of a coset the same phase.

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$$\begin{aligned} W_\Gamma(g_a) &\equiv 0 \pmod 8, \\ W_\Gamma(g_a \wedge g_b) &\equiv 0 \pmod 4, \\ W_\Gamma(g_a \wedge g_b \wedge g_c) &\equiv [\{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}] \pmod 2. \end{aligned}$$

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The cube satisfies all three: rows and faces have $W_\Gamma = 0$, edges have $W_\Gamma = 0$, and $u_1u_2u_3$ has weight 1.

From magic-state distillation to triorthogonal codes

2005 Bravyi–Kitaev [BK05]: magic-state distillation with ideal Clifford gates.

2012 Bravyi–Haah [BH12]: triorthogonal matrices, transversal T up to Clifford corrections.

2017 Campbell–Howard [CH17]: distillation with multiqubit gate synthesis, including CCZ.

2022 Nezami–Haah [NH22]: classify triorthogonal codes with $n + k \leq 38$ via RM indicators.

2024 Camps-Moreno, López, Matthews et al. [CLM+24]: CSS-T codes from cyclic and extended cyclic codes. In 2026 [CLM+26]: all diagonal transversal gates of a CSS code.

[BK05] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, Phys. Rev. A 71, 022316 (2005), arXiv:quant-ph/0403025.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

[CLM+24] E. Camps-Moreno et al., An algebraic characterization of binary CSS-T codes and cyclic CSS-T codes for quantum fault tolerance, Quantum Inf. Process. 23, 230 (2024), arXiv:2312.17518.

[CLM+26] E. Camps-Moreno et al., Transversal gates for quantum CSS codes, arXiv:2601.21514 (2026).

A triorthogonal matrix and a triorthogonal space differ

Matrix G [BH12]

$$\mathrm{wt}(g_a \wedge g_b) \equiv 0 \pmod 2,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv 0 \pmod 2.$$

Indices are distinct. Odd-weight rows label logical T outputs. Even-weight rows give X checks.

Space $\mathcal{H}$ [NH22]

$$\sum_{j=1}^{L} u_j v_j w_j = 0 \pmod 2$$

for all $u, v, w \in \mathcal{H} \subseteq \mathbb{F}_2^L$, including repeated vectors.

Unital: $\mathbf{1}_L \in \mathcal{H}$. Every word in $\mathcal{H}$ has even weight.

Nezami–Haah classify parent spaces, then obtain codes by puncturing and deleting coordinates.

CCZ needs an odd logical triple. Triorthogonal matrices have even distinct-row triples.

[BH12] S. Bravyi and J. Haah, Magic-state distillation with low overhead, Phys. Rev. A 86, 052329 (2012), arXiv:1209.2426.
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The nine CH conditions

K1	$\mathrm{wt}(K_i) \equiv 0$
K2	$\mathrm{wt}(K_i \wedge K_j) \equiv 0$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) \equiv 1$
KS1	$\mathrm{wt}(K_i \wedge S_l) \equiv 0$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) \equiv 0$
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S1	$\mathrm{wt}(S_l) \equiv 0$
S2	$\mathrm{wt}(S_l \wedge S_m) \equiv 0$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) \equiv 0$

All mod 2, indices distinct.

Reduce the three row congruences mod 2. Since Γ_j is odd, $W_\Gamma \equiv \mathrm{wt} \pmod 2$, and the result is these nine parities [CH17].

They give CCZ up to a diagonal Clifford correction: *quasi-transversal* CCZ. The cube satisfies all nine.

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A physical $T/T^\dagger$ layer with no correction needs the signed congruences mod 8 and mod 4. These are conditions on Γ as well as on G.

The CH conditions are necessary. A code can satisfy all nine and admit no sign vector Γ.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

Result: a 48-qubit code with a transversal $T/T^\dagger$ CCZ

The construction [LB26]

$Q_{48} = [\![48, 3, 3]\!]$, with $(d_X, d_Z) = (16, 3)$. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ with no Clifford correction.

- The stabilizer code is fixed by projective geometry: $\mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2)$.
- The logical rows come from a SAT search, and the signs from a linear system over $\mathbb{Z}_4$.
- Lower bound: no code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$. Lengths 39 to 46 are excluded by [WJS26].
- The same construction gives codes at 13 block lengths from 48 to 496.

The code satisfies the error-correction constraints and the gate constraints at once.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

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4. Solve for the signs Γ over $\mathbb{Z}_4$.

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Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \implies \text{weight-one errors are detected}, \qquad \beta_i \neq \beta_j \implies \beta_i \oplus \beta_j \neq 0.$$

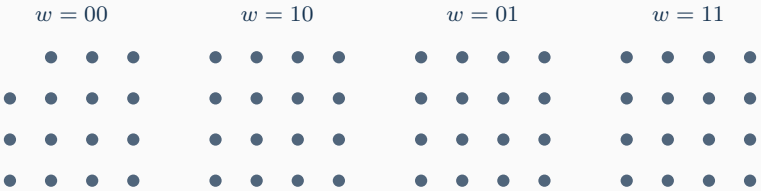
Distinct nonzero columns are points of $\mathrm{PG}(s-1,2)$, the $2^s - 1$ nonzero vectors of $\mathbb{F}_2^s$.

$d_Z \geq 3$ follows from the stabilizer columns alone, for any logical rows.

The support is projective space minus a subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

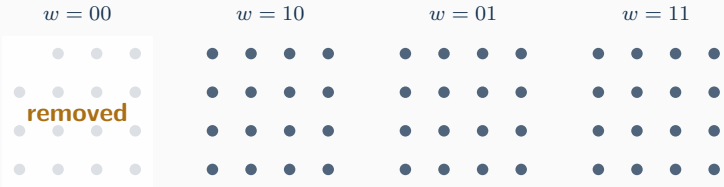
Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$. The removed $\mathrm{PG}(3,2)$ is the set of 15 nonzero points with $w = 00$.



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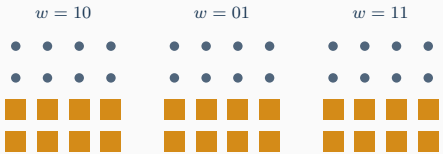


The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ are the rows of S.

Three blocks of 16 points give a projective $[48, 6]$ stabilizer code.

Hyperplane sections give the two nonzero weights

A word of $\mathcal{C}_2$ evaluates $\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w$ on $\mathcal{P}$, and $\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|$.



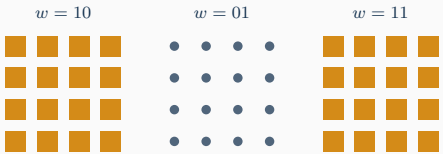
$\lambda \neq 0$, for example $\ell = u_1$: eight points in each block, weight $3 \cdot 8 = 24$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

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$\lambda = 0$, for example $\ell = w_1$: two full blocks, weight $2 \cdot 16 = 32$.

Projective two-weight codes correspond to finite-geometry sets with two hyperplane intersection sizes [CK86].

Weights 24 and 32 are multiples of 8, so S1–S3 hold.

[CK86] A. R. Calderbank and W. M. Kantor, The geometry of two-weight codes, Bull. London Math. Soc. 18, 97 (1986), doi:10.1112/blms/18.2.97.

Column indicators record the generator matrix

Let M have distinct columns in $\mathbb{F}_2^r$. Its *column indicator* [NH22] is the Boolean function

$$\chi_M(v) = 1 \iff v \text{ is a column of } M.$$

Write χ_M as a multilinear polynomial. Then $\chi_M \in \mathrm{RM}(\deg \chi_M, r)$ and $\mathrm{wt}(\chi_M)$ is the number of columns.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

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Cube: the columns of $G = [K; S]$ are $(u, 1)$ for $u \in \mathbb{F}_2^3$. In coordinates $(x, y) \in \mathbb{F}_2^3 \times \mathbb{F}_2$,

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$$\chi_G(x, y) = y.$$

Q_{48}: the columns of S are the points with $w \neq 00$, so

$$\chi_S(u, w) = 1 + (1 + w_1)(1 + w_2) = w_1 + w_2 + w_1 w_2.$$

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

The CH conditions are inner products with monomials

For Boolean functions on $\mathbb{F}_2^r$, let $\langle f, g\rangle = \sum_v f(v)g(v) \bmod 2$ [CH17]. Index the rows of G by coordinates $z = (x, y)$. For a set T of rows, $m_T = \prod_{t\in T} z_t$ and

$$\langle \chi_G, m_T\rangle = \mathrm{wt}\Big(\bigwedge_{t\in T} g_t\Big) \bmod 2.$$

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

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The nine CH conditions say: $\langle \chi_G, x_1x_2x_3\rangle = 1$, and $\langle \chi_G, m\rangle = 0$ for every other monomial of degree 1 to 3.

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Since $\mathrm{RM}(3, r)^\perp = \mathrm{RM}(r-4, r)$ and $y_1 \cdots y_s$ is one solution [LB26],

$$\chi_G \in y_1 \cdots y_s + \varepsilon\delta_0 + \mathrm{RM}(r-4, r), \qquad r = 3 + s,\ \varepsilon = n \bmod 2.$$

The cube indicator $\chi_G = y_1$ is this representative with $s = 1$.

[CH17] E. T. Campbell and M. Howard, Unified framework for magic state distillation and multiqubit gate synthesis with reduced resource cost, Phys. Rev. A 95, 022316 (2017), arXiv:1606.01904.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

The label map turns logical rows into Boolean unknowns

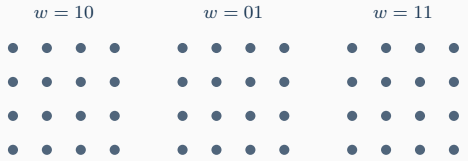
Column $\beta = (u, w) \in \mathcal{P}$ of S carries a label $\kappa(\beta) \in \mathbb{F}_2^3$, the column of K above it [LB26]:

$$\chi_G(x, u, w) = \chi_S(u, w) \prod_{i=1}^{3} \big(1 + x_i + \kappa_i(u, w)\big).$$

The CH conditions become

$$\bigoplus_{\beta \in \mathcal{P}} m_{T_K}\big(\kappa(\beta)\big)\, m_{T_S}(\beta) = [T = \{K_1, K_2, K_3\}].$$

S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



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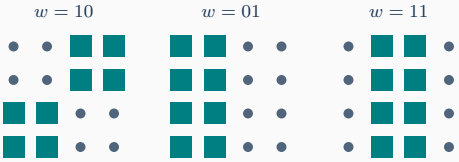
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S1–S3 have $T_K = \emptyset$ and hold by geometry. The rest have degree at most 3 in the label bits.



Teal: points with $\kappa_1 = 1$, the support of K_1.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

SAT assigns three logical bits to each geometric point

Unknowns: $\kappa(\beta) \in \mathbb{F}_2^3$ for the 48 points, so 144 label bits.

$$7 + 6s + 3\binom{s}{2} = 88 \text{ parity constraints at } s = 6.$$

K1–K3 give 7, KS1–KS2 give $6s$, and KS3 gives $3\binom{s}{2}$. Each is an XOR of products of up to three label bits. There are no distance constraints.

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A solution for Q_{48} [LB26]:

w	10	01	11
$\kappa_1(u, w)$	$u_1 \oplus u_3$	$1 \oplus u_3$	u_4

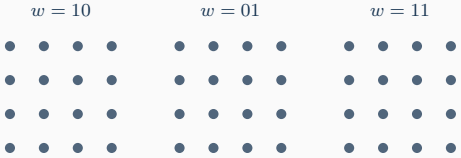
κ_2 and κ_3 are quadratic in u on each block. The rows must also be independent modulo $\mathcal{C}_2$.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

The signs solve a linear system over $\mathbb{Z}_4$

Write $\Gamma_j = 1 - 2\gamma_j$ with $\gamma_j \in \{0, 1\}$. Then $W_\Gamma(c) = \mathrm{wt}(c) - 2\,\gamma \cdot c$, and every codeword $c = xK \oplus yS$ has even weight:

$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$



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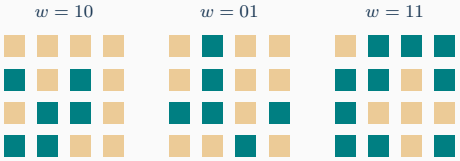
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$$\gamma \cdot c \equiv \tfrac{1}{2}\,\mathrm{wt}(c) - 2x_1x_2x_3 \pmod 4.$$

For Q_{48} [LB26]:

$$\gamma(u, w) = u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u),$$
$$Q_1 = u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4,$$
$$Q_2 = u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4.$$

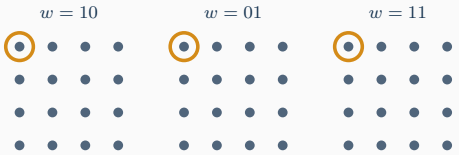


Teal: $T^\dagger$ $(6, 6, 10)$. Gold: T. Total $26\,T + 22\,T^\dagger$.

The pattern satisfies the gate condition on all $8 \cdot 64 = 512$ codewords.

[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

A three-point witness makes the Z-distance exact



Put a Z error at $u = 0000$ in each block:

$$\underbrace{10 \oplus 01 \oplus 11}_{\text{stabilizer syndrome}} = 00, \qquad \underbrace{011 \oplus 110 \oplus 000}_{\text{label sum}} = 101 \neq 000.$$

The operator commutes with every check and acts nontrivially on the encoded qubits.

Projectivity gives $d_Z \geq 3$. The witness gives $d_Z \leq 3$. $Q_{48} = [\![48, 3, 3]\!]$ with $(d_X, d_Z) = (16, 3)$.

Stabilizer supports are Reed–Muller words

For projective S with support $\mathcal{P} \subset \mathbb{F}_2^s$, S1–S3 say

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0 \pmod 2, \qquad 1 \le t \le 3.$$

Following [NH22], let $c = \chi_{\mathcal{P}} + \epsilon\delta_0$, with ϵ chosen so that $\mathrm{wt}(c)$ is even. The origin changes no moment of positive degree, so $c \perp \mathrm{RM}(3, s)$ and

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s-4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

For Q_{48}: $c = w_1 + w_2 + w_1 w_2 \in \mathrm{RM}(2, 6)$.

A short code needs a low-weight Reed–Muller word, then compatible labels and signs.

[NH22] S. Nezami and J. Haah, Classification of small triorthogonal codes, Phys. Rev. A 106, 012437 (2022), arXiv:2107.09684.

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Concurrent classification closes the 39–46 gap

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	[WJS26]
$n = 47$	Quasi-transversal CCZ	[JVB+26]
$n = 48$	Physical $T/T^\dagger$, no correction	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$, up to CNOT+S output equivalence. The nine CH conditions select their CCZ output class. Their classification does not test sign feasibility for every code.

[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).

What the 47-qubit comparison establishes

Jacinto et al. [JVB+26] give a $[\![47, 3, 3]\!]$ code satisfying the nine CH conditions. Its $\mathbb{Z}_4$ sign system is inconsistent [LB26], so it has no physical $T/T^\dagger$ layer, and its correction needs entangling CZ gates.

Wills–Jain–Singh [WJS26] establish minimum length 47 in the quasi-transversal CCZ class with $k = 3$, $d_Z \geq 3$. Their catalogue identifies outputs up to CNOT and S, not full Clifford equivalence.

These facts do not establish uniqueness of the length-47 code or rule out every other length-47 completion with feasible signs.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).
[LB26] B. Lu and K. R. Brown, Building codes with transversal CCZ using projective geometry and SAT solvers, arXiv:2610.09341 (2026).

Code length and distillation footprint are different resources

Apply the phase gates to encoded $|+\rangle^{\otimes 3}$, measure checks, and decode to prepare a CCZ resource state.

- n: code length, the number of noisy T inputs in the distillation description.
- N: circuit footprint in the implementation of [JVB+26]. For the matrix description, $N = k + s$ [WJS26].
- The 47-to-1 distance-3 CCZ protocol has $(n, N) = (47, 9)$.

At least 3 input Z errors are needed for an undetected logical error in the ideal-Clifford distillation model. A physical factory also needs a circuit, state preparation, measurements, decoding, and a noise model.

[JVB+26] H. Jacinto et al., Exploring the landscape of compact magic-state distillation factories, arXiv:2606.07734 (2026).
[WJS26] A. Wills, S. P. Jain, and S. Singh, Classification of generalised triorthogonal codes through length 54, arXiv:2609.30860 (2026).

The geometric construction produces a family

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

Each support needs compatible logical rows and physical signs. Existence does not follow from the point count.

The next algebraic task is to characterize which supports admit the required labels and signs.

Open questions connect algebra to implementation

- Characterize geometric supports admitting compatible logical rows and physical signs.
- Construct higher-distance families, other non-Clifford gates, and more logical qubits.
- Determine the minimum length for physical $T/T^\dagger$ CCZ without a correction.
- Compare full circuits, noise, and resource costs before claiming a practical advantage.

Paper
arXiv:2610.09341

Code and data
github.com/
jerrylvx/
gate2code-public

Geometry gives distance and divisibility. Labels and signed weights give the gate.

Joint work with Ken Brown. Thanks to Robert Calderbank for classical coding-theory pointers.